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### Calculation:

Weight of the organic substance taken = w g

Increase in weight of $H_2O$ = x g

Increase in weight of $CO_2$ = y g

18 g of $H_2O$ contain 2g of hydrogen

$\therefore x$ g of $H_2O$ contain $\left(\frac{2}{18} \times \frac{x}{w}\right)$ g of hydrogen

Percentage of hydrogen $= \left(\frac{2}{18} \times \frac{x}{w} \times 100\right)\%$

44g of $CO_2$ contains 12g of carbon

$\therefore y$ g of $CO_2$ contain $\left(\frac{12}{44} \times \frac{y}{w}\right)$ g of carbon

Percentage of Carbon $= \left(\frac{12}{44} \times \frac{y}{w} \times 100\right)\%$

### Note:

1. If the organic substance under investigation also contain N, it will produce oxides of nitrogen on combustion. A spiral of copper is introduced at the combustion tube, to reduce the oxides of nitrogen to nitrogen which escapes unabsorbed.

2. If the compound contains halogen as well, a spiral of silver is also introduced in the combustion tube. It converts halogen into silver halide.

3. In case if the substance also contains sulphur, the copper oxide in the combustion tube is replaced by lead chromate. The $SO_2$ formed during combustion is thus converted to lead sulphate and prevented from passing into the absorption unit.

### Worked out example: 1

0.26g of an organic compound gave 0.039 g of water and 0.245 g of carbon dioxide on combustion. Calculate the percentage of C & H

Weight of organic compound = 0.26g

Weight of water = 0.039g

Weight of $CO_2$ = 0.245g

**Percentage of hydrogen**

18 g of water contain 2 g of hydrogen

0.039 g of water contain $\frac{2}{18} \times \frac{0.039}{0.26}$

% of hydrogen = $\frac{0.039}{0.26} \times \frac{2}{18} \times 100 = 1.66\%$

**Percentage of carbon**

44 g of $CO_2$ contain 12 g of C

0.245 g of $CO_2$ contains $\frac{12}{44} \times \frac{0.245}{0.26}$ g of C

% of Carbon = $\frac{12}{44} \times \frac{0.245}{0.26} \times 100 = 25.69\ \%$

### Evaluate Yourself

5) 0.2346g of an organic compound containing C, H & O, on combustion gives 0.2754g of $H_2O$ and 0.4488g $CO_2$. Calculate the % composition of C, H & O in the organic compound [C=52.17, H = 13.04, O = 34.79]

### Estimation of sulphur:

**Carius method:** A known mass of the organic substance is heated strongly with fuming $HNO_3$. C & H get oxidized to $CO_2$& $H_2O$ while sulphur is oxidized to sulphuric acid as per the following reaction.